

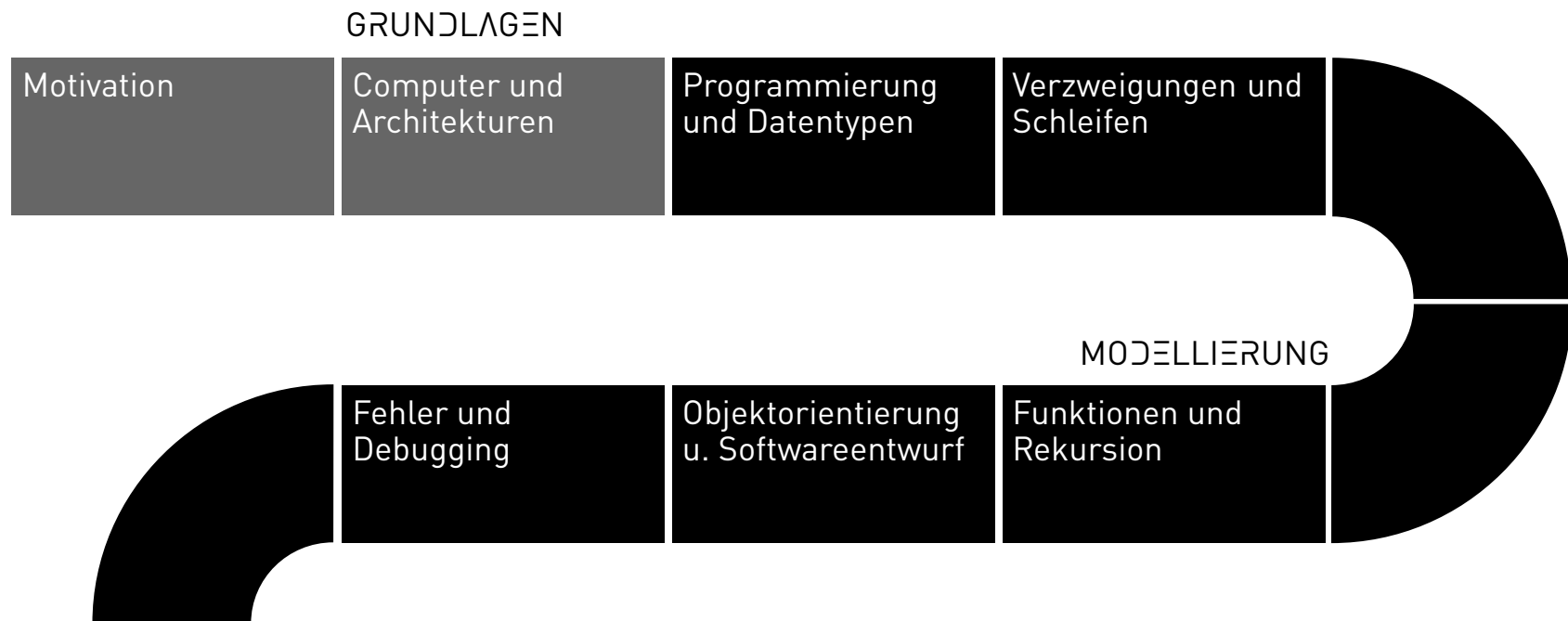
Programming and Databases

Joern Ploennigs

Computer hardware

MIDJOURNEY: MAN COMPUTER, REF. FRITZ KAI IN

PROCESS



DEFINITION: COMPUTER

Computers are ubiquitous nowadays and perform many tasks in our daily lives. They are formally defined as



Definition: Computer

A computer is a device that processes data using programmable calculation rules.

LECTURE HALL QUESTION

What types of computers are there?



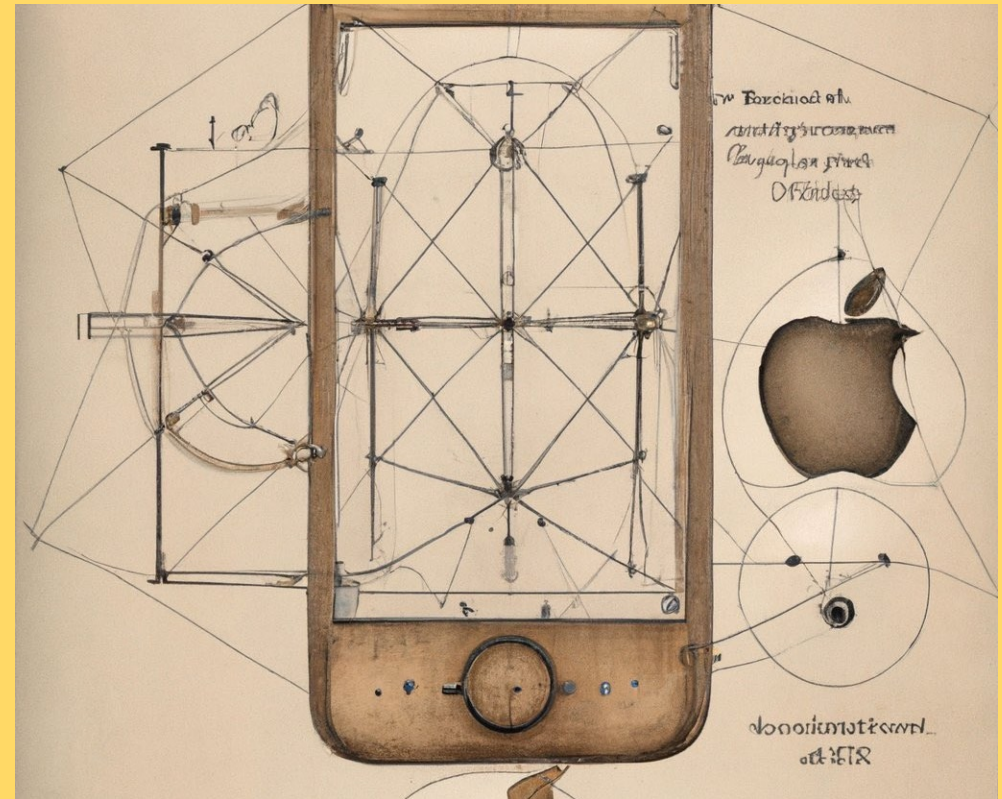
Midjourney: A Mona Lisa-style painting with a goat taking a photo with an iPad

TYPES OF COMPUTERS

- Supercomputer – Special-purpose computers with very many CPUs and GPUs for highly parallel processing of complex problems (e.g., weather simulations)
- Mainframe – Special-purpose large machines with very high reliability, e.g., in banks
- Server – Computers in data centers without a display for the Internet or cloud computing
- Personal Computer (PC) – Desktop computers in offices or at home for work, gaming, etc.
- Laptops – Mobile computers on the go for work, gaming, studying
- Smartphones – Mobile devices with a touchscreen and limited phone functionality
- Tablets – Mobile computers with large touchscreens and no phone functionality
- Embedded Computers – Small computers in cars, robots, and smart homes

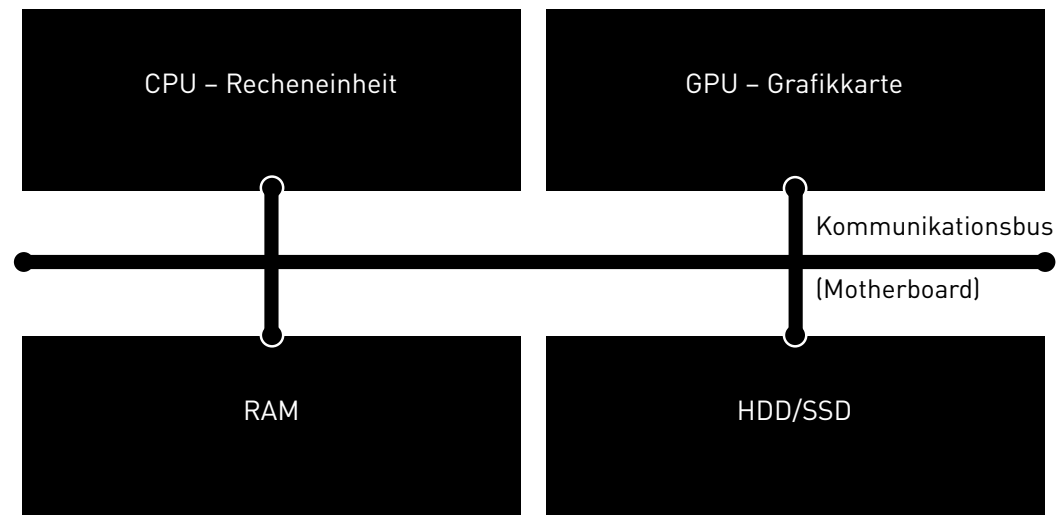
LECTURE HALL QUESTION

What hardware makes up a computer?



DALL-E 2: Early designs of the iPhone by Leonardo da Vinci

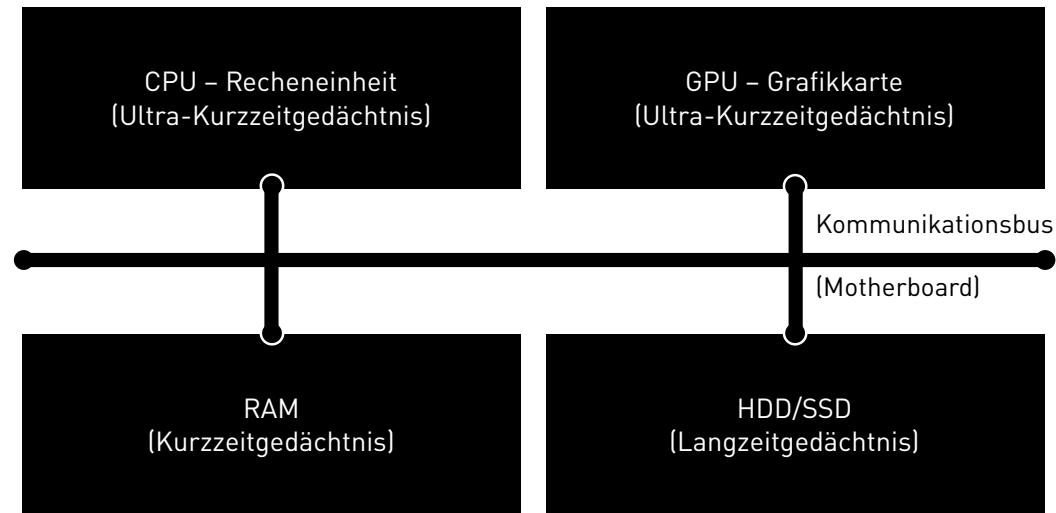
PC BUILD



The core components of a computer are:

- the CPU as the central processing unit,
- the GPU for graphical applications,
- the RAM as the main memory, and
- a hard drive (HDD) or solid-state drive (SSD)

COMPUTER ARCHITECTURE: STORAGE DURATION

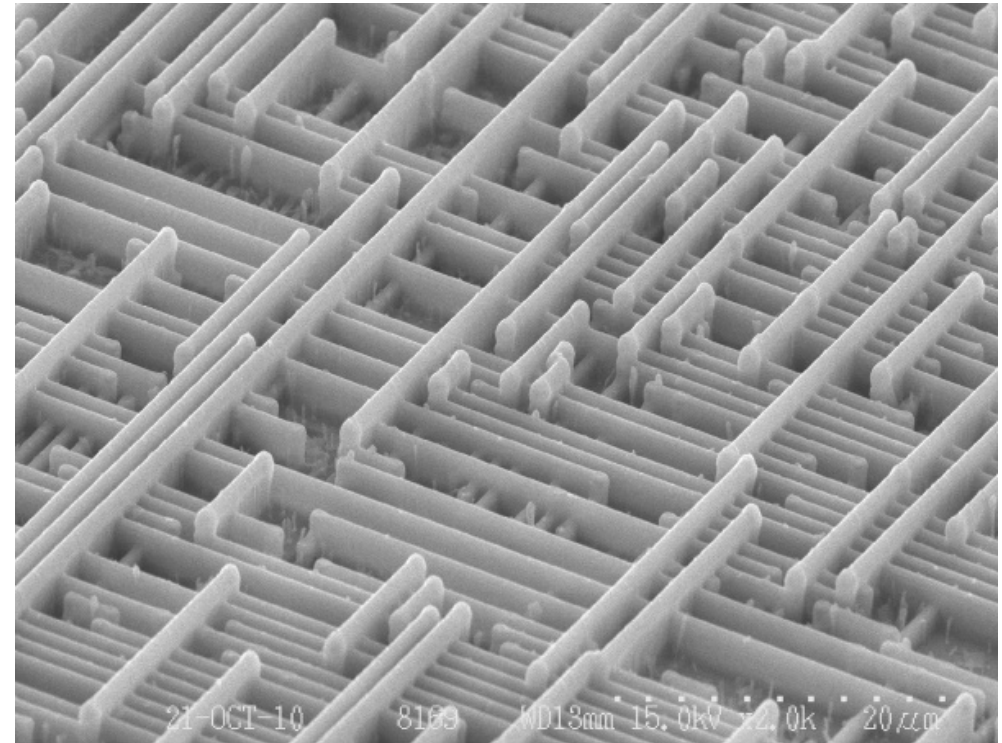


The computer has memory types similar to those of humans:

- CPUs and GPUs have small registers and cache memory (ultra-short-term memory)
- RAM is volatile memory, i.e., its contents are lost when powered off (short-term memory)
- The HDD/SSD is non-volatile storage, i.e., its contents remain (long-term memory)

CPU – CENTRAL PROCESSING UNIT

- The CPU is the computer's processing unit that processes data
- It processes a sequence of instructions (a program)
- A CPU consists of millions of transistors that are only a few nanometers in size (1000000nm=1cm)
- A few transistors can perform only one logical operation



<https://www.ial-fa.com/blog/wafer-lot-acceptance-to-mil-std-883>

BINARY NUMBERS

- Computers store, process & communicate data as binary numbers, because transistors in a gate can only perform logical operations.
- binarius – dual, double
- The smallest possible (useful) system of characters
- Possible representations of binary code:
 - 0 / 1
 - False / True
 - Off / On

NOTE: THE DIFFERENCE IN COMPUTING

Due to the binary representation, common unit prefixes such as kilo-, mega-, etc. are defined based on 1024 rather than 1000.

Unit	Abbreviation	Decimal Value	Binary (IEC)	Abbreviation	Binary (Bytes)
Kilobyte	kB	1,000	Kibibyte	KiB	1,024
Megabyte	MB	1,000,000	Mebibyte	MiB	1,048,576
Gigabyte	GB	1,000 ³	Gibibyte	GiB	1,024 ³
Terabyte	TB	1,000 ⁴	Tebibyte	TiB	1,024 ⁴

LECTURE HALL QUESTION

How far can you count with ten fingers?



Midjourney: Counting with two hands

BINARY NUMBERS - INTEGERS

- Our decimal numbers can be encoded as binary numbers
- Binary numbers allow the same well-known basic arithmetic operations as decimal numbers, namely addition, subtraction, multiplication, and division
- This is how the computer can perform calculations with decimal numbers
- In computers, numbers are also often encoded in hexadecimal (base 16), since a byte (8 bits) can be represented by two characters

Decimal Binary Hexadecimal

0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

BINARY NUMBERS – FLOATING-POINT

- Real numbers cannot be represented exactly in a computer, because they can have infinitely many decimal digits (e.g., $1/3$, π)
- Computers therefore use approximate floating-point numbers in exponential notation: $\pm m \times 2^e$
- Decomposition: sign bit, exponent (with bias), mantissa (binary)

⚠ **Attention:** Limited number of digits → rounding errors!

Example: We want to represent the number -13.25 as a floating-point number in the format $\pm m \times 2^e$:

1. *Sign:* The number is negative, so 1.
2. *Decimal number in binary form:*
 $13.25_{10} = 1101.01_2$
3. *Exponent shift:* $1101.01_2 = 1.10101_2 \times 2^3$
4. *Decomposition* into mantissa: 1.10101_2 and exponent: 3
5. *Result:* $-13.25 = -1.10101_2 \times 2^3$

In the IEEE 754 standard the bits are padded with zeros so that they fit into 32- or 64-bit representations.

- Sign bit: 1 (negative)
- Exponent with bias: $3 + 127 = 130_{10}$ (10000010₂)
- Mantissa padded with zeros: 101010000000000000000000
- The binary representation in memory is therefore: **1 10000010 101010000000000000000000**

CHARACTER REPRESENTATION IN COMPUTERS

- Characters in text are encoded in binary in computers
- ASCII: Text characters are stored in 8 binary values (8 bits = 1 byte)
- Binary encoding extends from the lowest hardware level up to everyday applications

Char	Decimal	Binary	Char	Decimal	Binary	Char	Decimal	Binary
a	97	01100001	A	65	01000001	0	48	00110000
b	98	01100010	B	66	01000010	1	49	00110001
c	99	01100011	C	67	01000011	2	50	00110010
d	100	01100100	D	68	01000100	3	51	00110011
e	101	01100101	E	69	01000101	4	52	00110100
f	102	01100110	F	70	01000110	5	53	00110101
g	103	01100111	G	71	01000111	6	54	00110110
h	104	01101000	H	72	01001000	7	55	00110111
i	105	01101001	I	73	01001001	8	56	00111000
j	106	01101010	J	74	01001010	9	57	00111001
k	107	01101011	K	75	01001011	!	33	00100001
l	108	01101100	L	76	01001100	?	63	00111111
m	109	01101101	M	77	01001101	.	46	00101110
n	110	01101110	N	78	01001110	,	44	00101100
o	111	01101111	O	79	01001111	Space	32	00100000

IMAGE REPRESENTATION IN COMPUTERS

- Images are also stored in binary form in a computer
- In this process, an image is divided into pixels (resolution)
- The color in each pixel is then stored as a number, e.g., true color RGB (16,777,216 color variations):
 - Blue – 0 ... 256 (= 1 Byte)
 - Green – 0 ... 256 (= 1 Byte)
 - Red – 0 ... 256 (= 1 Byte)
- The numbers are then encoded in binary, e.g., as 24 bits (1 Byte = 8 Bits; $3 \times 8 \text{ Bits} = 24 \text{ Bits}$)



Mahmut Sinecen: Digital Image Processing with MATLAB, 2015

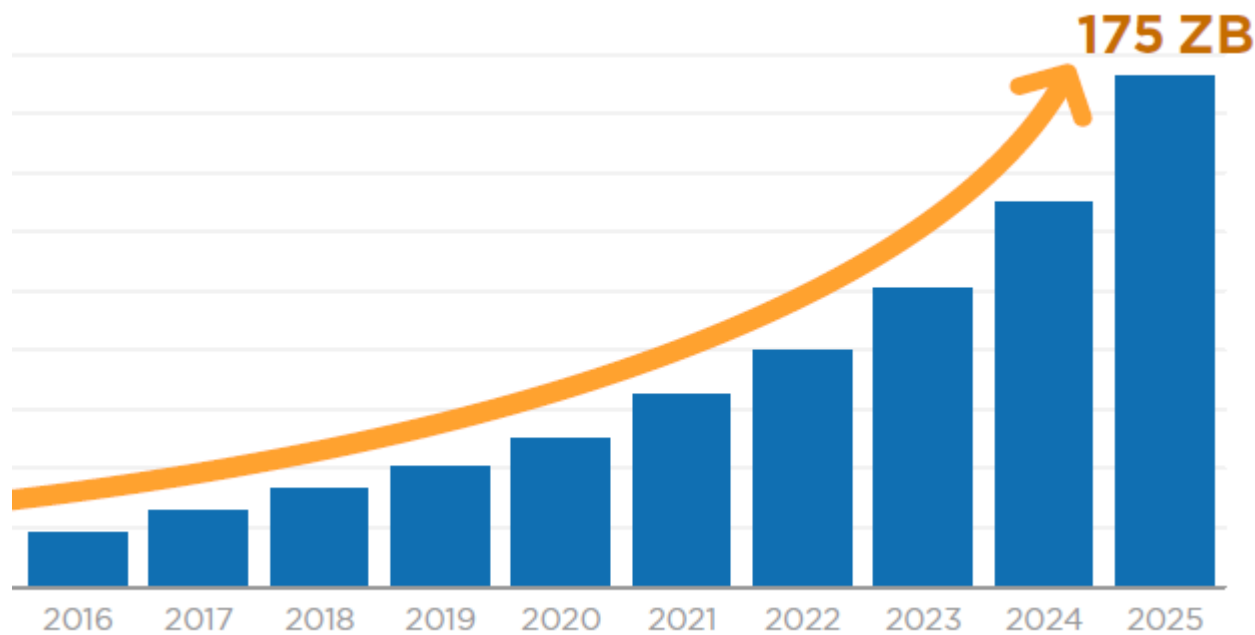
ATTENTION: THE DIFFERENCE IN COMPUTING

Due to the binary representation, common unit prefixes such as kilo-, mega-, etc. are defined on the basis of 1024 and not on the basis of 1000

Unit	Binary Name	Decimal (base-1000)	Decimal, written out	Binary (base-1024)	Binary, written out
Kilobyte	Kibibyte	1 KB = 10^3 B	1,000 B	1 KiB = 2^{10} B	1,024 B
Megabyte	Mebibyte	1 MB = 10^6 B	1,000,000 B	1 MiB = 2^{20} B	1,048,576 B
Gigabyte	Gibibyte	1 GB = 10^9 B	1,000,000,000 B	1 GiB = 2^{30} B	1,073,741,824 B
Terabyte	Tebibyte	1 TB = 10^{12} B	1,000,000,000,000 B	1 TiB = 2^{40} B	1,099,511,627,776 B

DATA GROWTH

- The amount of data generated is increasing.
- In the future you will face large amounts of data.
- The computer must help you analyze this data using computer science



Worldwide IDC Global
DataSphere Forecast, 2022–
2026

LECTURE HALL QUESTION

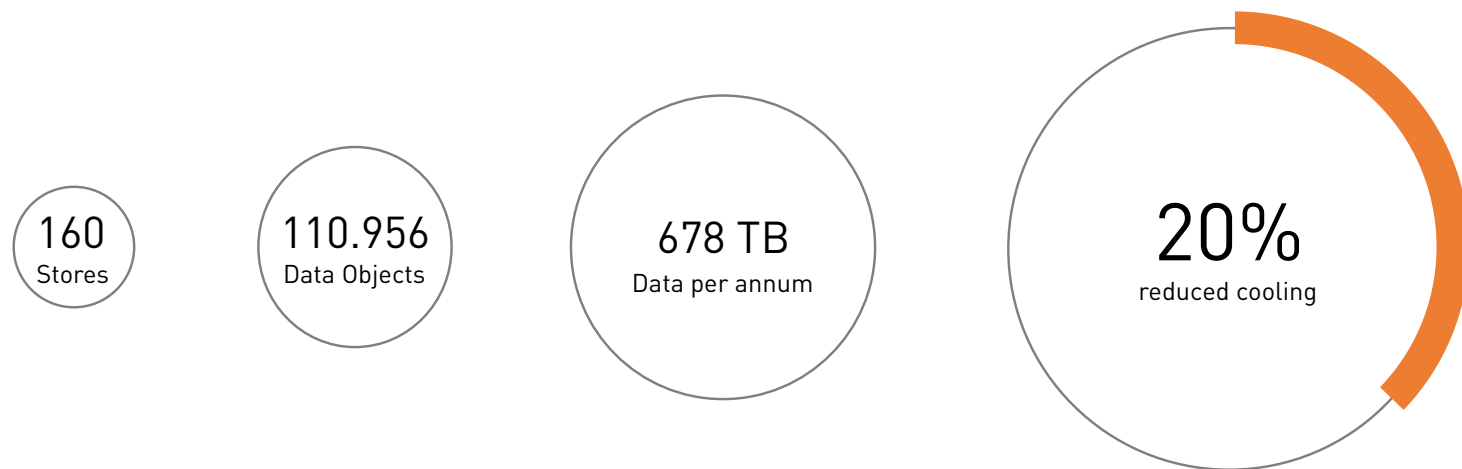
How many sensors does a supermarket have?



Midjourney: An alien supermarket filled with exotic food

EXAMPLE – TESCO IRELAND

- The supermarket chain Tesco invested early in sensors and monitoring systems across its stores with the goal of saving energy.
- The collected data grew so quickly and to such a scale that no one could analyze it.
- Using machine learning models, IBM Research was able to analyze the data and helped them cut cooling demand by 20%.



LESSONS LEARNED

- "Lessons Learned" are short off-topic digressions that I wish someone had told me about when I was a student
- Usually in the middle of lectures to lighten things up
- This semester has focused primarily on soft-skills topics; next semester, survival tips for the professional world



DALL-E 2: Teddy bears mixing sparkling chemicals as mad scientists, steampunk

LESSON LEARNED

What types of memory are you familiar with?



DALL-E 2: 1980s illustration of a face with a visible brain that's exploding

LESSONS LEARNED - MEMORY

- Knowledge takes time to consolidate.
- Tap into your subconscious, i.e., gather information and let it mature.
- Interval learning is, just as in sports, the most effective way to learn something in the long term.

Questions?

